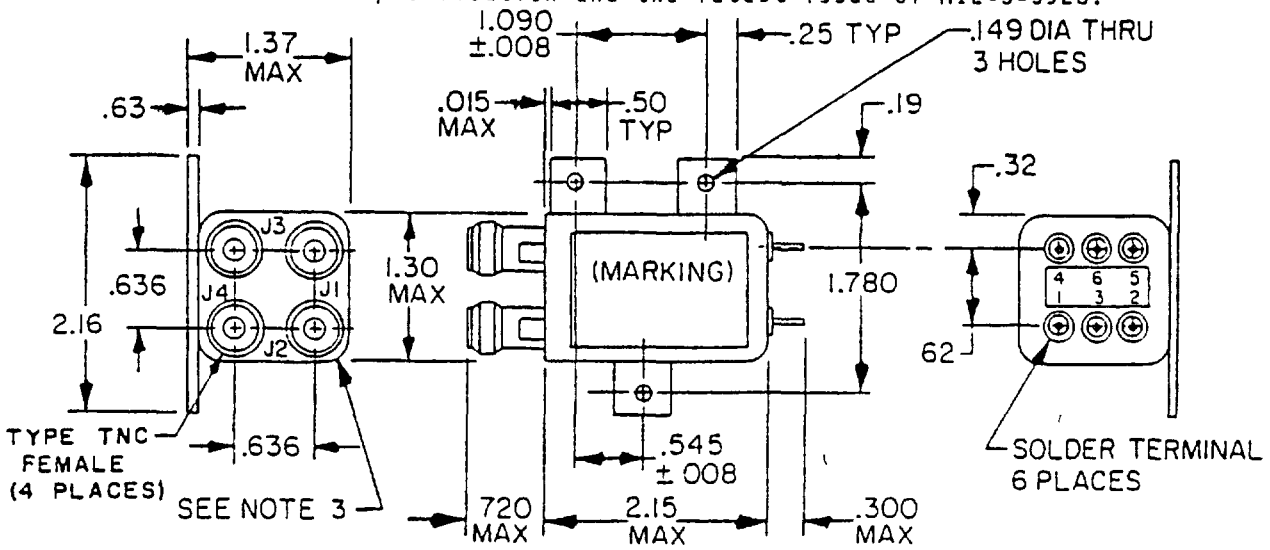
MILITARY SPECIFICATION SHEET

SWITCHES, RADIO FREQUENCY TRANSMISSION LINE (COAXIAL) (ELECTRICALLY OPERATED) CLASS 4, TR

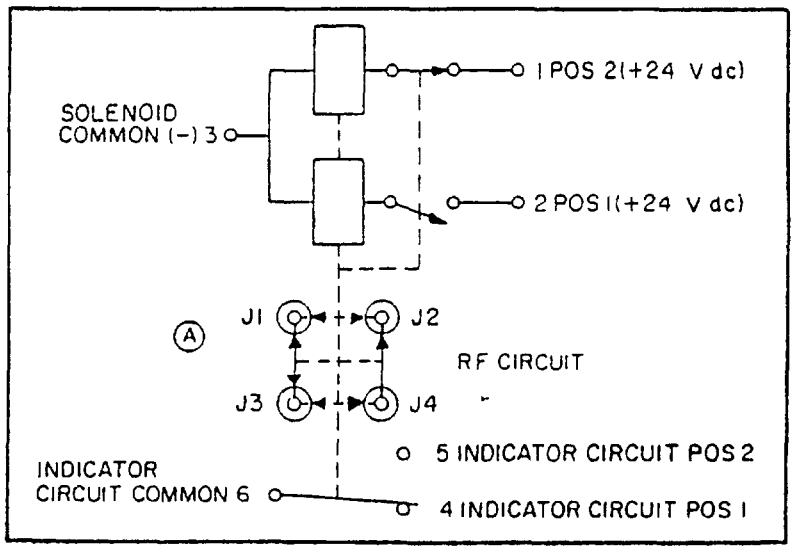
Ⓑ

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the switches described herein shall consist of this specification and the latest issue of MIL-S-3928.



Inches	mm
.008	0.20
.015	0 38
.149	3.78
.19	4.8
.25	6 4
.300	7.62
.32	8.1
.50	12.7
.545	13.84
.62	15.7
.63	16.0
.636	16.15
.720	18.29
1.090	27.69
1.30	33.0
1.37	32.8
1.780	45.21
2.15	54 61
2.16	54.86



POSITION 1 ENERGIZED TERMINAL NO 2 J1-J3 AND J2-J4
POSITION 2 ENERGIZED TERMINAL NO 1 J1-J2 AND J3-J4 (DOTTED LINES)

SCHEMATIC (SHOWN IN POSITION 1)

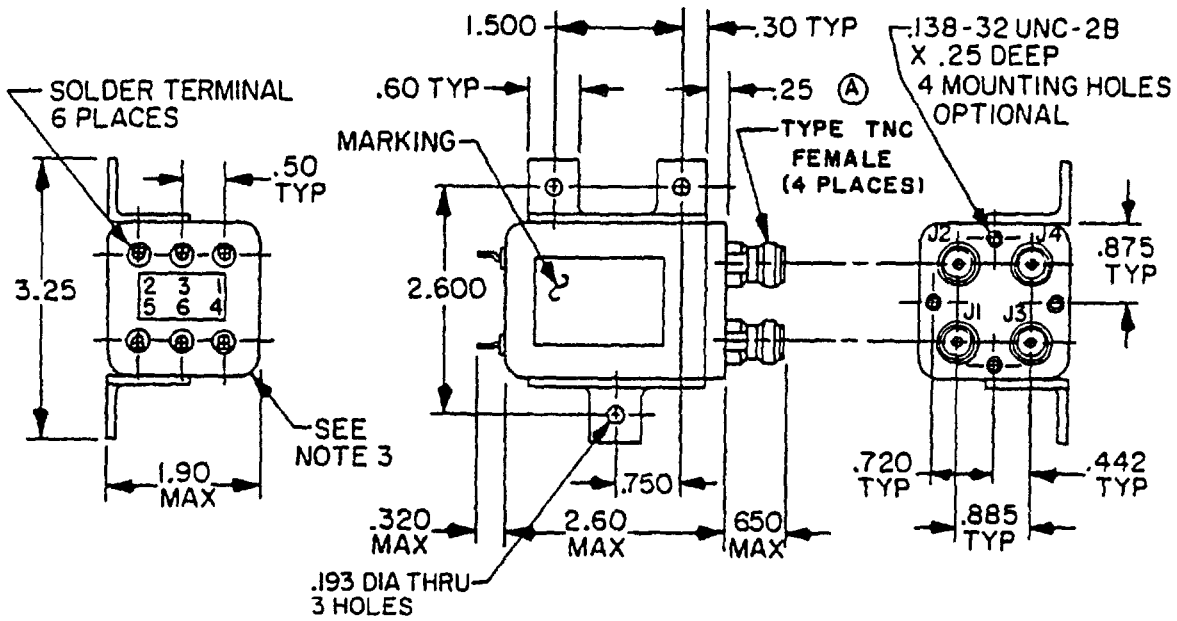
NOTES
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances are ±.010 (0.25 mm) for three place decimals and ±.03 (0.8 mm) for two place decimals.
4. Round corners of case may be squared.

FIGURE 1. Part number M3928/21-01.

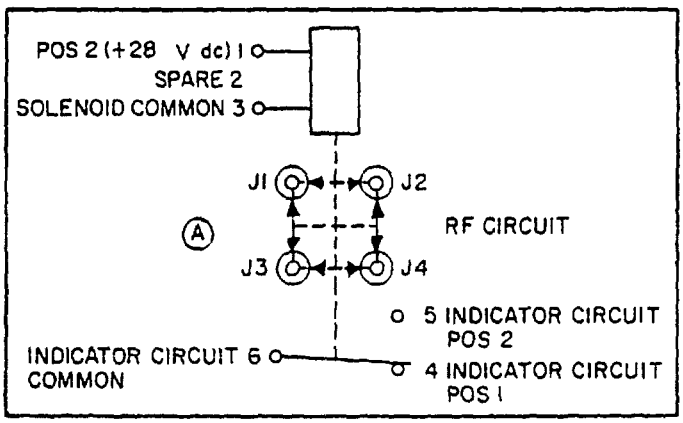
Ⓑ denotes changes

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Inches	mm
.138	3.51
.193	4.90
.25	6.4
.30	7.6
.320	8.13
.442	11.23
.50	12.7
.60	15.2
.650	16.51
.720	18.29
.750	19.05
.875	22.23
.885	22.48
1.500	38.10
1.90	48.3
2.600	66.04
3.25	82.6



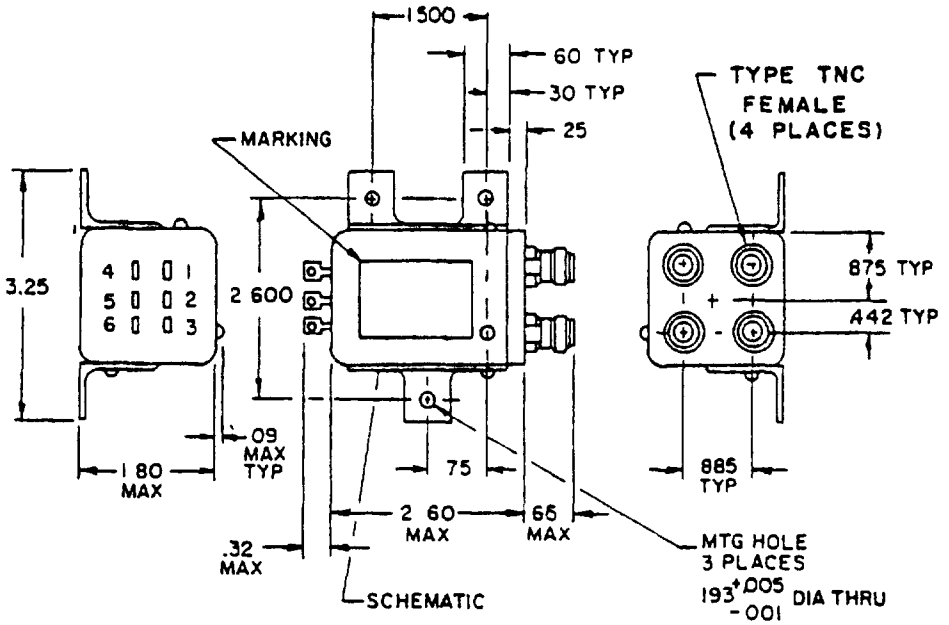
POSITION 1 DEENERGIZED OR FAILSAFE POSITION, J1 - J3 AND J2 - J4
POSITION 2 ENERGIZED POSITION, J1 - J2 AND J3 - J4.

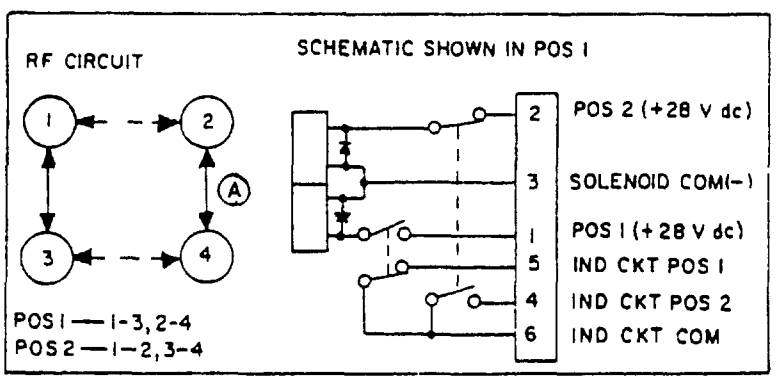
SCHEMATIC (SHOWN IN POSITION 1)

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances are ±.010 (0.25 mm) for three place decimals and ±.03 (0.8 mm) for two place decimals.
4. Round corners of case may be squared.

FIGURE 2. Part number M3928/21-02.

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Inches	mm	Inches	mm
.001	0.03	.65	16.5
.005	0.13	.75	19.1
.09	2.3	.875	22.23
.193	4.90	.885	22.48
.25	6.4	1.500	38.10
.30	7.6	1.80	45.7
.32	8.1	2.600	66.04
.442	11.23	3.25	82.6
.60	15.2		

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances are ±.010 (0.25 mm) for three place decimals and ±.03 (0.8 mm) for two place decimals.
4. Round corners of case may be squared.

FIGURE 3. Part number M3928/21-03.

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Ⓑ TABLE I Electrical and performance characteristics

Part no M3928/21-	Figure no	Housing	Frequency range dc to GHz	VSWR	Insertion loss (dB)	Isolation (dB)	Switching time (ms)	Position indication circuit and rating	Life cycles x 1000	Fail-safe or latching	Operating current (A) 1/	Holding current (A) 1/	Nominal operating voltage	Pickup voltage (less than)	Dropout voltage (less than)	Power and indicator connector	Weight (oz)
				Max	Max	Min	Max				Max	Max					Max
2/ 01N,S	1	I	11	1 5 1	0 5	50	30	Yes	100	L	2.6	0	28 V dc	20 V dc	---	Solder terminals	6
02N,S	2	E	12 4	1 5 1	0 5	60	25	2A @ 28 V dc	100	F	0 5	0 2	28 V dc	20 V dc	5 V dc	Solder terminals	12
03N,S	3	I	12 4 1 57	1 5 1 1.2 1	0 5 0 1	60	20 30	0 1A @ 28 V dc	100	F	0 65	---	28 V dc	20 V dc	---	Solder terminals	12

1/ At 28 V dc and 20°C

2/ Transient interference (RFI).

3/ RF power handling capacity. 80 watts minimum at sea level and 10^{-5} through 10^{-10} torr

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REQUIREMENTS:

Dimensions and configurations: See figures 1, 2, and 3.

RF connectors Female connectors (4 places) shall meet the requirements of MIL-C-39012 and shall mate with TNC type male connectors per MIL-C-39012/26.

Electrical and performance characteristics. See table I.

RF power handling capability (average): 50 watts (minimum).

Nominal impedance: 50 ohms.

Termination: Open.

Vibration: Method I.

Operating temperature: -55°C to +85°C.

Part number: M3928/21- (and dash number from table I).

(B) NOTE. Switches covered by this specification sheet are substitutable for the manufacturer's or original equipment manufacturer's (OEM) part numbers specified in table II. This information in no way implies that the manufacturer's or OEM part number or NSN is suitable as a substitute for the military part number.

TABLE II. Substitution data.

Part number M3928/21-	OEM or manufacturer's code	OEM or manufacturer's part number	NSN 5985-
01	07397	31-807221	00-139-1745
01	82152	700C30200	00-139-1745
02	28528	1131516G1	00-630-6674
02	82152	310C30800	00-630-6674
03	28528	1133650G1	---
03	82152	300C30200	---
03	24022	CS-47T6B	---

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - MI
Navy - OS
Air Force - 11, 99
DLA - ES

User activities:
Army - AV
Navy - AS, MC, SH
Air Force - 19

Preparing activity:
Navy - EC

Agent:
DLA - ES

(Project 5985-1016-13)

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